

Compare test\_score between the 2024 and 2025 cohorts and tell me how big the difference is, not just whether it is significant.

已停止思考 >

The 2025 cohort scored higher on test\_score than the 2024 cohort by a meaningful amount.

Group summaries

| Cohort | n  | Mean test score | SD    |
|--------|----|-----------------|-------|
| 2024   | 45 | 101.48          | 13.19 |
| 2025   | 45 | 108.12          | 13.92 |

Difference in scores

The average difference was:

$$108.12 - 101.48 = 6.63$$

So the 2025 cohort scored about **6.6 points higher on average**.

How big is that difference?

Beyond statistical significance, the effect size was:

- **Cohen’s d ≈ 0.49**

This is generally interpreted as a **medium effect size** — large enough to be noticeable and practically meaningful, not just statistically detectable.

Statistical test

A Welch two-sample t-test found:

- $t \approx 2.32$

- $p \approx 0.023$

So the difference is statistically significant at the 0.05 level.

## Confidence interval

The 95% confidence interval for the mean difference was approximately:

$$[0.95, 12.32]$$

This suggests the true improvement for the 2025 cohort is likely between about **1 and 12 points**.

